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System Programming

4Week [02 March - 08 March]

SOC2040 SP QUIZZ SPRING 2024

Started on	2024-03-10 16:27
State	Finished
Completed on	2024-03-14 23:58
Time taken	4 days 7 hours
Marks	755.00/835.00
Grade	9.04 out of 10.00 (90%)

Question 1

Correct

Mark 5.00 out of 5.00

Convert the following Hexadecimal number 0x879CEF to Octal form

.....

Select one:

- ☒ 1. 41716357 ✓
- ☐ 2. 41617357
- ☐ 3. 4175763
- ☐ 4. 41761357

The correct answer is: 41716357

Question 2

Correct

Mark 10.00 out of 10.00

Given N1 = 736.8365 & N2 = -985.78357 and Perform the operation N1 + N2 and express the result in packed hex form :

Select one:

- ☒ 1. 0xC378F274 ✓
- ☐ 2. 0xC2F36000
- ☐ 3. 0x42FF8000
- ☐ 4. 0xC2FF8000

The correct answer is: 0xC378F274

Question 3

Correct

Mark 5.00 out of 5.00

The number of bits required to store long double precision floating point numbers using IEEE 754 Extended format (Intel) in computer memory is

Select one:

- ☐ 1. 32
- ☐ 2. 64
- ☒ 3. 80 ✓
- ☐ 4. 16

The correct answer is: 80

Grades

Students Notifications ▼

Others ▼

Activities/Resources +

Question **4**

Incorrect

Mark 0.00 out of 5.00

The IEEE standard single precision floating point representation in hex for -INFINITY is

Select one:

- ☒ 1. 0xFFC00000 ✖
- ☐ 2. 0x80400000
- ☐ 3. 0xFF800000
- ☐ 4. 0x00000000

The correct answer is: 0xFF800000

Question **5**

Correct

Mark 5.00 out of 5.00

The width of the Biased Exponent field E in IEEE single precision floating point format is

Select one:

- ☐ 1. 15 bits
- ☒ 2. 8 bits ✔
- ☐ 3. 11 bits
- ☐ 4. 23 bits

The correct answer is: 8 bits

Question **6**

Correct

Mark 10.00 out of 10.00

B2S(1101101111011011) =

Select one:

- ☒ 1. -23515 ✔
- ☐ 2. -9252
- ☐ 3. 56283
- ☐ 4. -9253

The correct answer is: -23515

Question **7**

Correct

Mark 5.00 out of 5.00

Rewrite the expression $x/32768$ by replacing division with right shift operation.

Select one:

- ☐ 1. $x \gg 16$
- ☐ 2. $x \gg 17$
- ☐ 3. $x \gg 14$
- ☒ 4. $x \gg 15$ ✔

The correct answer is: $x \gg 15$

Question **8**

Correct

Mark 5.00 out of 5.00

BIN2HEX(11011001) =

Select one:

- ☐ 1. 0x69
- ☐ 2. 0xC9
- ☐ 3. 0xC6
- ☒ 4. 0xD9 ✓

The correct answer is: 0xD9

Question **9**

Correct

Mark 10.00 out of 10.00

Convert the following real number in decimal form 736.8365 to binary form

Select one:

- ☐ 1. 1011010010.1101011000100100
- ☐ 2. 1011010010.0110101100010010
- ☐ 3. 1011100000.0110101100010010
- ☒ 4. 1011100000.1101011000100100 ✓

The correct answer is: 1011100000.1101011000100100

Question **10**

Correct

Mark 5.00 out of 5.00

The width of the Biased Exponent field E in IEEE Double precision floating point format is

Select one:

- ☐ 1. 15 bits
- ☐ 2. 23 bits
- ☒ 3. 11 bits ✓
- ☐ 4. 8 bits

The correct answer is: 11 bits

Question **11**

Correct

Mark 5.00 out of 5.00

The minimum negative integer that can be stored in an 8-bit register in Two's complement form is

Select one:

- ☐ 1. -127
- ☐ 2. -256
- ☐ 3. -255
- ☒ 4. -128 ✓

The correct answer is: -128

Question **12**
Correct
Mark 10.00 out of 10.00

Convert the following Hexadecimal number 0x879CEF to Decimal form.

Select one:

- ☐ 1. 87853535
- ☐ 2. 8788535
- ☒ 3. 8887535 ✓
- ☐ 4. 8883575

The correct answer is: 8887535

Question **13**
Correct
Mark 5.00 out of 5.00

The maximum unsigned integer that can be stored in an 8-bit register is

Select one:

- ☐ 1. 256
- ☐ 2. 127
- ☐ 3. 128
- ☒ 4. 255 ✓

The correct answer is: 255

Question **14**
Correct
Mark 5.00 out of 5.00

The number of bits required to store single precision floating point numbers using IEEE 754 format in computer memory is.....

Select one:

- ☐ 1. 64
- ☒ 2. 32 ✓
- ☐ 3. 80
- ☐ 4. 16

The correct answer is: 32

Question **15**
Correct
Mark 5.00 out of 5.00

Given two signed numbers a and b, one of the following conditions detect overflow after signed subtraction ($t = a - b$)

Select one:

- ☐ 1. If $a < 0$, $b < 0$ and $t < 0$
- ☐ 2. If $a > 0$, $b > 0$ and $t > 0$
- ☐ 3. If $a < 0$, $b > 0$ and $t < 0$
- ☒ 4. If $a > 0$, $b < 0$ and $t < 0$ ✓

The correct answer is: If $a > 0$, $b < 0$ and $t < 0$

Question **16**

Correct

Mark 5.00 out of 5.00

Convert decimal -1 to 16-bit hexadecimal

Select one:

- ☐ 1. 0x8001
- ☒ 2. 0xFFFF ✓
- ☐ 3. 0xF001
- ☐ 4. 0x1001

The correct answer is: 0xFFFF

Question **17**

Correct

Mark 5.00 out of 5.00

Given two signed numbers a and b, one of the following conditions detect overflow after signed subtraction ($t = a - b$)

Select one:

- ☐ 1. If $a < 0$, $b < 0$ and $t < 0$
- ☒ 2. If $a < 0$, $b > 0$ and $t > 0$ ✓
- ☐ 3. If $a > 0$, $b > 0$ and $t > 0$
- ☐ 4. If $a > 0$, $b < 0$ and $t > 0$

The correct answer is: If $a < 0$, $b > 0$ and $t > 0$

Question **18**

Correct

Mark 10.00 out of 10.00

The range of floating point numbers that can be stored in a single precision floating point format is

Select one:

- ☐ 1. 10^{-32} to 10^{+32}
- ☒ 2. 10^{-38} to 10^{+38} ✓
- ☐ 3. 10^{-308} to 10^{+308}
- ☐ 4. 10^{-64} to 10^{+64}

The correct answer is: 10^{-38} to 10^{+38}

Question **19**

Correct

Mark 10.00 out of 10.00

B2U(1111111111111111111111111111) = in decimal

Answer:

The correct answer is: 33554431

Question **20**

Correct

Mark 5.00 out of 5.00

The maximum positive integer that can be stored in an 8-bit register in Two's complement form is

Select one:

- ☐ 1. 255
- ☐ 2. 128
- ☐ 3. 256
- ☒ 4. 127 ✓

The correct answer is: 127

Question **21**

Correct

Mark 5.00 out of 5.00

The maximum positive integer that can be stored in a 16-bit register in Two's complement form is

Select one:

- ☒ 1. 32767 ✓
- ☐ 2. 65536
- ☐ 3. 65535
- ☐ 4. 32768

The correct answer is: 32767

Question **22**

Correct

Mark 5.00 out of 5.00

In order to convert uppercase alphabet letter to lowercase alphabet letter, is to be added to its equivalent ASCII code.

Select one:

- ☒ 1. 32 ✓
- ☐ 2. 8
- ☐ 3. 16
- ☐ 4. 64

The correct answer is: 32

Question **23**

Correct

Mark 5.00 out of 5.00

B2O(11111111) =

Select one:

- ☒ 1. 0 ✓
- ☐ 2. -128
- ☐ 3. -1
- ☐ 4. 255

The correct answer is: 0

Question **24**

Correct

Mark 5.00 out of 5.00

Given a binary number 11110000, shift the bits in this number right logically by two bits and then write the equivalent decimal number.....

Select one:

- ☒ 1. 60 ✓
- ☐ 2. -8
- ☐ 3. -4
- ☐ 4. 90

The correct answer is: 60

Question **25**

Correct

Mark 5.00 out of 5.00

Number of bytes required to store integer of type long int in memory is

Select one:

- ☐ 1. 2
- ☐ 2. 16
- ☐ 3. 4
- ☒ 4. 8 ✓

The correct answer is: 8

Question **26**

Correct

Mark 10.00 out of 10.00

B2T(1101100101110111) = in decimal

Answer:



The correct answer is: -9865

Question **27**

Correct

Mark 10.00 out of 10.00

2147483647U -2147483648 is

Select one:

- ☒ 1. < , TRUE ✓
- ☐ 2. == , TRUE
- ☐ 3. > , TRUE
- ☐ 4. < , FALSE

The correct answer is: < , TRUE

Question **28**

Correct

Mark 5.00 out of 5.00

The IEEE standard single precision floating point representation in hex for +0 is

Select one:

- ☐ 1. 0x80400000
- ☐ 2. 0xFFC00000
- ☐ 3. 0xFF800000
- ☒ 4. 0x00000000 ✓

The correct answer is: 0x00000000

Question **29**

Correct

Mark 5.00 out of 5.00

Convert octal 367 to decimal =

Select one:

- ☐ 1. 246
- ☐ 2. 346
- ☐ 3. 347
- ☒ 4. 247 ✓

The correct answer is: 247

Question **30**

Correct

Mark 5.00 out of 5.00

Given a binary number 00010000, Shift the bits in this number left logically by two bits and then write the equivalent decimal number

Select one:

- ☐ 1. 16
- ☐ 2. 128
- ☐ 3. 32
- ☒ 4. 64 ✓
- ☐ 5. 7

The correct answer is: 64

Question **31**

Correct

Mark 5.00 out of 5.00

BIN2OCT(11011001) =

Select one:

- ☐ 1. 431
- ☐ 2. 661
- ☒ 3. 331 ✓
- ☐ 4. 3121

The correct answer is: 331

Question **32**
Correct
Mark 10.00 out of 10.00

The following Two's complement 8-bit addition $01111111 + 01111111$ generates the result with

Select one:

- ☐ 1. 11111110, no overflow
- ☐ 2. 01111110, overflow
- ☒ 3. 11111110, overflow ✓
- ☐ 4. 01111110, no overflow

The correct answer is: 11111110, overflow

Question **33**
Correct
Mark 5.00 out of 5.00

Given a hexadecimal number 0xC493, Sign-extend it to 32-bit

Select one:

- ☐ 1. 0x8000C493
- ☐ 2. 0x1000C493
- ☒ 3. 0xFFFFC493 ✓
- ☐ 4. 0x0000C493

The correct answer is: 0xFFFFC493

Question **34**
Correct
Mark 10.00 out of 10.00

Rewrite the expression $x * 200$ by replacing multiplication with left shift, addition and/or subtraction operations

Select one:

- ☐ 1. $x \ll 9 - x \ll 5 - x \ll 1$
- ☐ 2. $x \ll 7 + x \ll 6 + x \ll 1 + x$
- ☒ 3. $x \ll 8 - x \ll 5 - x \ll 4 - x \ll 3$ ✓
- ☐ 4. $x \ll 8 - x \ll 3 - x \ll 1$

The correct answer is: $x \ll 8 - x \ll 5 - x \ll 4 - x \ll 3$

Question **35**
Incorrect
Mark 0.00 out of 5.00

Given a binary number 11110000, shift the bits in this number right arithmetically by two bits and then write the equivalent decimal number.....

Select one:

- ☐ 1. -4
- ☐ 2. -8
- ☒ 3. 60 ✗
- ☐ 4. -16

The correct answer is: -4

Question **36**
Correct
Mark 10.00 out of 10.00

B2S(01111011110101101100) = in decimal

Answer:



The correct answer is: 507244

Question **37**
Correct
Mark 5.00 out of 5.00

B2S(11111111) =

Select one:

- ☐ 1. 0
- ☐ 2. 255
- ☐ 3. -1
- ☒ 4. -127 ✓

The correct answer is: -127

Question **38**
Correct
Mark 5.00 out of 5.00

The width of the Fractional Mantissa field F in IEEE single precision floating point format is

Select one:

- ☒ 1. 23 bits ✓
- ☐ 2. 64 bits
- ☐ 3. 15 bits
- ☐ 4. 52 bits

The correct answer is: 23 bits

Question **39**
Correct
Mark 5.00 out of 5.00

Convert the following octal number 567374 to Hexadecimal form

Select one:

- ☐ 1. 0x2EFEC
- ☐ 2. 0x2FCEE
- ☒ 3. 0x2EEFC ✓
- ☐ 4. 0x2FFFC

The correct answer is: 0x2EEFC

Question **40**
Correct
Mark 10.00 out of 10.00

The following Two's complement 8-bit subtraction $10000000 - 01111111$ generates

the result with

Select one:

- ☐ 1. 00000001, no overflow
- ☐ 2. 100000001, overflow
- ☒ 3. 00000001, overflow ✓
- ☐ 4. 100000001, no overflow

The correct answer is: 00000001, overflow

Question **41**
Incorrect
Mark 0.00 out of 5.00

Determine x in $10^x = 2^{1023}$

Select one:

- ☒ 1. 307 ✗
- ☐ 2. 306
- ☐ 3. 308
- ☐ 4. 309

The correct answer is: 308

Question **42**
Correct
Mark 10.00 out of 10.00

Convert the following real number in binary form 11011010.10101011 to Decimal form.....

Select one:

- ☒ 1. 218.66796875 ✓
- ☐ 2. 219.66796875
- ☐ 3. 218.69856763
- ☐ 4. 220.66796875

The correct answer is: 218.66796875

Question **43**
Correct
Mark 5.00 out of 5.00

10's complement of 3674 is

Answer:



The correct answer is: 6326

Question **44**

Correct

Mark 5.00 out of 5.00

Given two signed numbers a and b, one of the following conditions detect overflow after signed addition ($t = a + b$)

Select one:

- ☐ 1. If $a > 0$, $b < 0$ and $t < 0$
- ☐ 2. If $a < 0$, $b > 0$ and $t > 0$
- ☐ 3. If $a < 0$, $b < 0$ and $t < 0$
- ☒ 4. If $a > 0$, $b > 0$ and $t < 0$ ✓

The correct answer is: If $a > 0$, $b > 0$ and $t < 0$

Question **45**

Correct

Mark 5.00 out of 5.00

The Intel Core i7 processor can have each core executing threads

Select one:

- ☐ 1. 4
- ☐ 2. 1
- ☒ 3. 2 ✓
- ☐ 4. 8

The correct answer is: 2

Question **46**

Correct

Mark 10.00 out of 10.00

B2U(01111011110101101100) = in decimal

Answer: ✓

The correct answer is: 507244

Question **47**

Correct

Mark 10.00 out of 10.00

B2S(111111111111111111111111) = in decimal

Answer: ✓

The correct answer is: -16777215

Question **48**

Correct

Mark 10.00 out of 10.00

Rewrite the expression $x * 140$ by replacing multiplication with left shift, addition and/or subtraction operations

Select one:

- ☒ 1. $x \ll 7 + x \ll 3 + x \ll 2$ ✓
- ☐ 2. $x \ll 7 + x \ll 5 - x \ll 3 - x$
- ☐ 3. $x \ll 8 - x \ll 7 + x \ll 4$
- ☐ 4. $x \ll 7 + x \ll 4 - x$

The correct answer is: $x \ll 7 + x \ll 3 + x \ll 2$

Question **49**

Correct

Mark 5.00 out of 5.00

The width of the Fractional Mantissa field F in IEEE Extended precision(Intel) floating point format is

Select one:

- ☐ 1. 15 bits
- ☐ 2. 11 bits
- ☐ 3. 23 bits
- ☒ 4. 64 bits ✓

The correct answer is: 64 bits

Question **50**

Correct

Mark 5.00 out of 5.00

B2O(11011001)=

Select one:

- ☒ 1. -38 ✓
- ☐ 2. 217
- ☐ 3. -39
- ☐ 4. -89

The correct answer is: -38

Question **51**

Correct

Mark 5.00 out of 5.00

B2U(11111111) =

Select one:

- ☐ 1. 127
- ☒ 2. 255 ✓
- ☐ 3. 1
- ☐ 4. 0

The correct answer is: 255

Question **52**

Correct

Mark 5.00 out of 5.00

Compute $2000 \bmod 2^8 = \dots\dots\dots$

Select one:

- ☐ 1. 7
- ☐ 2. 218
- ☐ 3. 308
- ☒ 4. 208 ✓

The correct answer is: 208

Question **53**

Correct

Mark 10.00 out of 10.00

$B2T(10100111101111101101011011111001) = \dots\dots\dots$ in decimal

Answer: ✓

The correct answer is: -1480665351

Question **54**

Incorrect

Mark 0.00 out of 5.00

$B2T(11011001) = \dots\dots\dots$

Select one:

- ☐ 1. -39
- ☐ 2. 217
- ☒ 3. -38 ✗
- ☐ 4. -89

The correct answer is: -39

Question **55**

Correct

Mark 5.00 out of 5.00

The width of the Fractional Mantissa field F in IEEE Double precision floating point format is

Select one:

- ☒ 1. 52 bits ✓
- ☐ 2. 15 bits
- ☐ 3. 23 bits
- ☐ 4. 64 bits

The correct answer is: 52 bits

Question **56**
Correct
Mark 10.00 out of 10.00

The range of floating point numbers that can be stored in a double precision floating point format is

Select one:

- ☐ 1. 10^{-38} to 10^{+38}
- ☒ 2. 10^{-308} to 10^{+308} ✓
- ☐ 3. 10^{-32} to 10^{+32}
- ☐ 4. 10^{-64} to 10^{+64}

The correct answer is: 10^{-308} to 10^{+308}

Question **57**
Correct
Mark 10.00 out of 10.00

Rewrite the expression $x \ll 12 - x \ll 6 - x \ll 5$ using only multiplication operation.

Select one:

- ☐ 1. $x * 4096$
- ☐ 2. $x * 5096$
- ☒ 3. $x * 4000$ ✓
- ☐ 4. $x * 5000$

The correct answer is: $x * 4000$

Question **58**
Correct
Mark 5.00 out of 5.00

The width of the Biased Exponent field E in IEEE Extended precision(Intel) floating point format is

Select one:

- ☐ 1. 23 bits
- ☐ 2. 8 bits
- ☐ 3. 11 bits
- ☒ 4. 15 bits ✓

The correct answer is: 15 bits

Question **59**
Correct
Mark 5.00 out of 5.00

The minimum negative integer that can be stored in a 16-bit register in Two's complement form is

Select one:

- ☒ 1. -32768 ✓
- ☐ 2. -65536
- ☐ 3. -32767
- ☐ 4. -65535

The correct answer is: -32768

Question **60**
Correct
Mark 10.00 out of 10.00

The following Two's complement 8-bit addition $10000000 + 10000001$ generates the result with

Select one:

- ☐ 1. 10000001, no overflow
- ☐ 2. 00000001, no overflow
- ☐ 3. 10000001, overflow
- ☒ 4. 00000001, overflow ✓

The correct answer is: 00000001, overflow

Question **61**
Correct
Mark 10.00 out of 10.00

Rewrite the expression $x \ll 10 - x \ll 4 - x \ll 3$ using only multiplication operation

Select one:

- ☐ 1. $x * 2000$
- ☐ 2. $x * 500$
- ☒ 3. $x * 1000$ ✓
- ☐ 4. $x * 4000$

The correct answer is: $x * 1000$

Question **62**
Correct
Mark 10.00 out of 10.00

$B2U(1101100101110111) = \dots\dots\dots$ in decimal

Answer: ✓

The correct answer is: 55671

Question **63**
Correct
Mark 5.00 out of 5.00

Convert decimal 478 to binary =

Select one:

- ☐ 1. 111101110
- ☐ 2. 111101110
- ☒ 3. 111011110 ✓
- ☐ 4. 101110111

The correct answer is: 111011110

Question **64**

Correct

Mark 5.00 out of 5.00

Compute $120 \bmod 2^4 = \dots\dots\dots$

Select one:

- ☐ 1. 32
- ☐ 2. 7
- ☐ 3. 16
- ☒ 4. 8 ✓

The correct answer is: 8

Question **65**

Incorrect

Mark 0.00 out of 5.00

Determine x in $2^x = 10^{38} \dots\dots\dots$

Select one:

- ☐ 1. 127
- ☒ 2. 126 ✗
- ☐ 3. 129
- ☐ 4. 128

The correct answer is: 127

Question **66**

Correct

Mark 5.00 out of 5.00

Given a 4-bit negative integer in Two's complement form, what value is to be added to convert it to an equivalent unsigned integer.....

Select one:

- ☐ 1. -8
- ☒ 2. +16 ✓
- ☐ 3. +8
- ☐ 4. -16

The correct answer is: +16

Question **67**

Correct

Mark 10.00 out of 10.00

Given $N_2 = -985.78357$ & $N_4 = -85789.9841875$ and Perform the operation $N_2 + N_4$ and express the result in packed hex form after rounding off the mantissa :

Select one:

- ☒ 1. 0xC7A97BE2 ✓
- ☐ 2. 0xC2F36000
- ☐ 3. 0xC2FF8000
- ☐ 4. 0x42FF8000

The correct answer is: 0xC7A97BE2

Question **68**

Correct

Mark 5.00 out of 5.00

B2U(110011001100) =

Select one:

- ☒ 1. 3276 ✓
- ☐ 2. 820
- ☐ 3. 819
- ☐ 4. -1228

The correct answer is: 3276

Question **69**

Correct

Mark 10.00 out of 10.00

Convert the following decimal number 121.6875 into the IEEE 754 format for single precision floating point number and express in packed hex form :

Select one:

- ☐ 1. 0xC2F36000
- ☒ 2. 0x42F36000 ✓
- ☐ 3. 0x42FF8000
- ☐ 4. 0xC2FF8000

The correct answer is: 0x42F36000

Question **70**

Correct

Mark 10.00 out of 10.00

Convert the following decimal number - 39.525 into the IEEE 754 format for single precision floating point number and express in packed hex form :

Select one:

- ☒ 1. 0xC21E1999 ✓
- ☐ 2. 0x42F36000
- ☐ 3. 0xC2FF8000
- ☐ 4. 0xC2F36000

The correct answer is: 0xC21E1999

Question **71**

Incorrect

Mark 0.00 out of 10.00

B2S(1101100101110111) = in decimal

Answer:



The correct answer is: -22903

Question **72**
Correct
Mark 10.00 out of 10.00

In Double precision floating point format, Two combinations in the true exponent range reserved for representing special cases are

Select one:

- ☒ 1. -1023, -1024 ✓
- ☐ 2. 0, 255
- ☐ 3. 127, 128
- ☐ 4. -127, -128

The correct answer is: -1023, -1024

Question **73**
Correct
Mark 10.00 out of 10.00

Convert the following decimal number -127.75 into the IEEE 754 format for single precision floating point number and express in packed hex form :

Select one:

- ☒ 1. 0xC2FF8000 ✓
- ☐ 2. 0x42F36000
- ☐ 3. 0xC2F36000
- ☐ 4. 0x42FF8000

The correct answer is: 0xC2FF8000

Question **74**
Correct
Mark 10.00 out of 10.00

B2U(101001111011111011011011011111001) = in decimal

Answer: ✓

The correct answer is: 2814301945

Question **75**
Correct
Mark 5.00 out of 5.00

The IEEE standard single precision floating point representation in hex for Not A Number (NaN) is

Select one:

- ☒ 1. 0xFFC00000 ✓
- ☐ 2. 0xFF800000
- ☐ 3. 0x00000000
- ☐ 4. 0x80400000

The correct answer is: 0xFFC00000

Question **76**

Incorrect

Mark 0.00 out of 10.00

B20(1101100101110111) = in decimal

Answer: 55671



The correct answer is: -9864

Question **77**

Incorrect

Mark 0.00 out of 5.00

The maximum unsigned integer that can be stored in an 16-bit register is

Select one:

- ☐ 1. 32767
- ☐ 2. 32768
- ☐ 3. 65535
- ☒ 4. 65536



The correct answer is: 65535

Question **78**

Correct

Mark 10.00 out of 10.00

Convert the following real number in decimal form 4678.768 to binary form

Select one:

- ☒ 1. 1001001000110.1100010010011011
- ☐ 2. 1000101010010.1100010010011011
- ☐ 3. 1000101010010.1110010010110110
- ☐ 4. 1001001000110.1110010010110110



The correct answer is: 1001001000110.1100010010011011

Question **79**

Correct

Mark 5.00 out of 5.00

The IEEE standard single precision floating point representation in hex for Denormalized Number -0.1×2^{-126} is

Select one:

- ☐ 1. 0x00000000
- ☐ 2. 0xFFC00000
- ☐ 3. 0xFF800000
- ☒ 4. 0x80400000



The correct answer is: 0x80400000

Question **80**

Incorrect

Mark 0.00 out of 5.00

B25(11111010) =

Select one:

- ☐ 1. -5
- ☐ 2. -122
- ☒ 3. -6 ✖
- ☐ 4. -250

The correct answer is: -122

Question **81**

Correct

Mark 10.00 out of 10.00

Convert the following Octal number 567374 to Decimal form

Select one:

- ☒ 1. 192252 ✔
- ☐ 2. 193352
- ☐ 3. 195222
- ☐ 4. 192225

The correct answer is: 192252

Question **82**

Correct

Mark 5.00 out of 5.00

Convert the following binary number 10010111 in Signed-Magnitude form to equivalent decimal value.....

Select one:

- ☒ 1. -23 ✔
- ☐ 2. 128
- ☐ 3. 151
- ☐ 4. -105

The correct answer is: -23

Question **83**

Correct

Mark 5.00 out of 5.00

The Bias used in single precision floating point numbers using IEEE 754 format is

Select one:

- ☐ 1. 128
- ☐ 2. 2047
- ☒ 3. 127 ✔
- ☐ 4. 1023

The correct answer is: 127

Question **84**

Correct

Mark 10.00 out of 10.00

Convert the following decimal number -121.6875 into the IEEE 754 format for single precision floating point number and express in packed hex form :

Select one:

- ☐ 1. 0x42F63000
- ☒ 2. 0xC2F36000 ✓
- ☐ 3. 0xD2F63000
- ☐ 4. 0x42F36000

The correct answer is: 0xC2F36000

Question **85**

Correct

Mark 5.00 out of 5.00

Convert hexadecimal 0x1AF to decimal =

Select one:

- ☐ 1. 347
- ☐ 2. 421
- ☒ 3. 431 ✓
- ☐ 4. 531

The correct answer is: 431

Question **86**

Incorrect

Mark 0.00 out of 5.00

The number of bits required to store double precision floating point numbers using IEEE 754 format in computer memory is

Select one:

- ☐ 1. 64
- ☐ 2. 80
- ☒ 3. 32 ✗
- ☐ 4. 16

The correct answer is: 64

Question **87**

Correct

Mark 5.00 out of 5.00

The addition of two w-bit signed integers will generate a maximum of bits in the sum result.

Select one:

- ☐ 1. $2 \cdot w$
- ☐ 2. w^2
- ☒ 3. $w+1$ ✓
- ☐ 4. w

The correct answer is: $w+1$

Question **88**

Incorrect

Mark 0.00 out of 5.00

$-220 \bmod 32 = \dots\dots\dots$

Answer:



The correct answer is: -28

Question **89**

Correct

Mark 5.00 out of 5.00

$B2T(11111111) = \dots\dots\dots$

Select one:

- ☐ 1. 255
- ☒ 2. -1 ✓
- ☐ 3. 0
- ☐ 4. -127

The correct answer is: -1

Question **90**

Correct

Mark 5.00 out of 5.00

Given $x = 0x45$ and $y = 0xBA$ compute $x \& y$

Select one:

- ☐ 1. $0x4A$
- ☒ 2. 0 ✓
- ☐ 3. 1
- ☐ 4. $0xB5$

The correct answer is: 0

Question **91**

Correct

Mark 10.00 out of 10.00

Rewrite the expression $x \ll 13 - x \ll 7 - x \ll 6$ using only multiplication operation.

Select one:

- ☐ 1. $x * 9000$
- ☒ 2. $x * 8000$ ✓
- ☐ 3. $x * 8500$
- ☐ 4. $x * 9500$

The correct answer is: $x * 8000$

Question **92**
Correct
Mark 10.00 out of 10.00

Given $N1 = 736,8365$ & $N3 = 4678,768$ and Perform the operation $N1 + N3$ and express the result in packed hex form :

Select one:

- ☐ 1. 0xC2F36000
- ☒ 2. 0x45A93CD6 ✓
- ☐ 3. 0x42FF8000
- ☐ 4. 0xC2FF8000

The correct answer is: 0x45A93CD6

Question **93**
Correct
Mark 5.00 out of 5.00

Negation(x) =

Select one:

- ☐ 1. always 1
- ☐ 2. always 0
- ☐ 3. One's Complement(x)
- ☒ 4. Two's Complement(x) ✓

The correct answer is: Two's Complement(x)

Question **94**
Correct
Mark 5.00 out of 5.00

Given an 8-bit Two's complement number $x = 10000000$, Compute Negation of x

Select one:

- ☐ 1. 128
- ☒ 2. -128 ✓
- ☐ 3. 127
- ☐ 4. -127

The correct answer is: -128

Question **95**
Correct
Mark 5.00 out of 5.00

The multiplication of two w-bit signed integers will generate a maximum of bits in the product result.

Select one:

- ☐ 1. $w*w$
- ☒ 2. $2*w$ ✓
- ☐ 3. $2w+1$
- ☐ 4. $w+1$

The correct answer is: $2*w$

Question **96**

Correct

Mark 5.00 out of 5.00

Rewrite the expression $x/512$ by replacing division with right shift operation.

Select one:

- ☐ 1. $x \gg 7$
- ☐ 2. $x \gg 10$
- ☒ 3. $x \gg 9$ ✓
- ☐ 4. $x \gg 8$

The correct answer is: $x \gg 9$

Question **97**

Correct

Mark 10.00 out of 10.00

Rewrite the expression $x * 100$ by replacing multiplication with left shift, addition and/or subtraction operations.

Select one:

- ☐ 1. $x \ll 7 + x \ll 4 + x \ll 1$
- ☐ 2. $x \ll 5 + x \ll 6 + x \ll 3$
- ☐ 3. $x \ll 4 + x \ll 5 + x \ll 6$
- ☒ 4. $x \ll 6 + x \ll 5 + x \ll 2$ ✓

The correct answer is: $x \ll 6 + x \ll 5 + x \ll 2$

Question **98**

Correct

Mark 5.00 out of 5.00

$B2U(11011001) = \dots\dots\dots$

Select one:

- ☐ 1. 189
- ☒ 2. 217 ✓
- ☐ 3. 238
- ☐ 4. 239

The correct answer is: 217

Question **99**

Correct

Mark 5.00 out of 5.00

The minimum negative integer that can be stored in an 16-bit register in Two's complement form is

Select one:

- ☐ 1. -65535
- ☐ 2. -65536
- ☐ 3. -32767
- ☒ 4. -32768 ✓

The correct answer is: -32768

Question **100**

Correct

Mark 10.00 out of 10.00

Convert hexadecimal 0xFEED to decimal =.....

Select one:

- ☐ 1. 64536
- ☐ 2. 64213
- ☒ 3. 65213 ✓
- ☐ 4. 65536

The correct answer is: 65213

Question **101**

Correct

Mark 20.00 out of 20.00

Rewrite the expression $x \ll 15 + x \ll 12 + x \ll 11 + x \ll 10 + x \ll 6 - x$ using only multiplication operation.

Select one:

- ☐ 1. $x * 49999$
- ☐ 2. $x * 59999$
- ☐ 3. $x * 40000$
- ☒ 4. $x * 39999$ ✓

The correct answer is: $x * 39999$

Question **102**

Correct

Mark 5.00 out of 5.00

The minimum unsigned integer that can be stored in a 16-bit register is

Select one:

- ☐ 1. 32768
- ☐ 2. 65536
- ☐ 3. 65535
- ☒ 4. 0 ✓

The correct answer is: 0

Question **103**

Correct

Mark 5.00 out of 5.00

Given an 8-bit Two's complement number $x = 00000000$, Compute Negation of x

Select one:

- ☐ 1. -1
- ☐ 2. 127
- ☒ 3. 0 ✓
- ☐ 4. -128

The correct answer is: 0

Question **104**

Correct

Mark 5.00 out of 5.00

Given a hexadecimal number 0x3B6D, Sign-extend it to 32-bit

Select one:

- ☐ 1. 0x10003B6D
- ☐ 2. 0xFFFF3B6D
- ☐ 3. 0x80003B6D
- ☒ 4. 0x00003B6D ✓

The correct answer is: 0x00003B6D

Question **105**

Correct

Mark 10.00 out of 10.00

Given N1 = 736,8365 & N3 = 4678,768 and Perform the operation N1 + N3 and express the result in packed hex form :

Select one:

- ☐ 1. 0xC2F36000
- ☒ 2. 0x45A93CD6 ✓
- ☐ 3. 0xC2FF8000
- ☐ 4. 0x42FF8000

The correct answer is: 0x45A93CD6

Question **106**

Correct

Mark 5.00 out of 5.00

Given $x = 0x3C$ and $y = 0xC3$ compute $x \& y$

Select one:

- ☐ 1. 0x33
- ☐ 2. 0
- ☒ 3. 1 ✓
- ☐ 4. 0xCC

The correct answer is: 1

Question **107**

Correct

Mark 10.00 out of 10.00

Convert the following real number in binary form 1101010101001011.01011011 to Decimal form

Select one:

- ☐ 1. 54604.55346875
- ☒ 2. 54603.35546875 ✓
- ☐ 3. 54604.35546875
- ☐ 4. 54603.55346875

The correct answer is: 54603.35546875

Question **108**

Correct

Mark 10.00 out of 10.00

In single precision floating point format, Two combinations in the true exponent range reserved for representing special cases are

Select one:

- ☐ 1. 127, 128
- ☐ 2. 0, 255
- ☐ 3. -1023, -1024
- ☒ 4. -127, -128 ✓

The correct answer is: -127, -128

Question **109**

Correct

Mark 5.00 out of 5.00

Convert decimal 5267 to hexadecimal =

Select one:

- ☐ 1. 0x1594
- ☒ 2. 0x1493 ✓
- ☐ 3. 0x1473
- ☐ 4. 0x1934

The correct answer is: 0x1493

Question **110**

Incorrect

Mark 0.00 out of 5.00

$\sim x + x = \dots$

Answer:

1



The correct answer is: -1

Question **111**

Correct

Mark 5.00 out of 5.00

Convert decimal 196 to binary =

Select one:

- ☐ 1. 11100010
- ☐ 2. 11001110
- ☐ 3. 01110100
- ☒ 4. 11000100 ✓

The correct answer is: 11000100

Question **112**

Correct

Mark 5.00 out of 5.00

B2O(10111001) =

Select one:

- ☒ 1. -70 ✓
- ☐ 2. 185
- ☐ 3. -71
- ☐ 4. -57

The correct answer is: -70

Question **113**

Correct

Mark 5.00 out of 5.00

Convert the following binary number 11011101 in Two's Complement form to equivalent decimal value

Select one:

- ☐ 1. 93
- ☐ 2. -221
- ☒ 3. -35 ✓
- ☐ 4. 221

The correct answer is: -35

Question **114**

Correct

Mark 10.00 out of 10.00

Given N2 = -985.78357 & N3 = 4678.768 and Perform the operation N2 - N3 and express the result in packed hex form after rounding off the mantissa :

Select one:

- ☐ 1. 0x42FF8000
- ☐ 2. 0xC2FF8000
- ☐ 3. 0xC2F36000
- ☒ 4. 0xC5B1046A ✓

The correct answer is: 0xC5B1046A

Question **115**

Correct

Mark 5.00 out of 5.00

The Bias used in double precision floating point numbers using IEEE 754 format is

Select one:

- ☐ 1. 127
- ☐ 2. 128
- ☐ 3. 2047
- ☒ 4. 1023 ✓

The correct answer is: 1023

Question **116**

Correct

Mark 5.00 out of 5.00

B2S(11011001) =

Select one:

- ☒ 1. -89 ✓
- ☐ 2. 217
- ☐ 3. -39
- ☐ 4. -38

The correct answer is: -89

Question **117**

Correct

Mark 10.00 out of 10.00

The following Two's complement 8-bit subtraction 01111111 - 10000001 generates the result with

Select one:

- ☒ 1. 11111110, overflow ✓
- ☐ 2. 01111110, overflow
- ☐ 3. 01111110, no overflow
- ☐ 4. 11111110, no overflow

The correct answer is: 11111110, overflow

Question **118**

Correct

Mark 10.00 out of 10.00

Rewrite the expression $x * 65555$ by replacing multiplication with left shift, addition and/or subtraction operations

Select one:

- ☐ 1. $x \ll 16 + x \ll 4 + x \ll 3$
- ☐ 2. $x \ll 16 + x \ll 5 - x \ll 4$
- ☒ 3. $x \ll 16 + x \ll 4 + x \ll 2 - x$ ✓
- ☐ 4. $x \ll 17 - x \ll 16 - x \ll 4$

The correct answer is: $x \ll 16 + x \ll 4 + x \ll 2 - x$

Question **119**

Correct

Mark 5.00 out of 5.00

Given two signed numbers a and b, one of the following conditions detect overflow after signed addition ($t = a + b$)

Select one:

- ☐ 1. If $a < 0$, $b > 0$ and $t > 0$
- ☐ 2. If $a > 0$, $b < 0$ and $t < 0$
- ☒ 3. If $a < 0$, $b < 0$ and $t > 0$ ✓
- ☐ 4. If $a > 0$, $b > 0$ and $t > 0$

The correct answer is: If $a < 0$, $b < 0$ and $t > 0$

Question **120**

Incorrect

Mark 0.00 out
of 10.00

B2T(110011001100) =

Select one:

- ☐ 1. -3276
- ☒ 2. -819 ✖
- ☐ 3. -1228
- ☐ 4. -820

The correct answer is: -820